

3. The diagram below shows an absorption spectrum of an object in space.



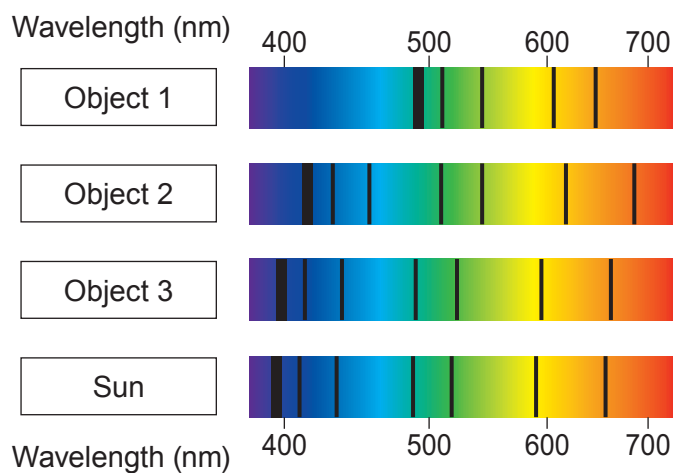
- (a) Complete the following sentences by underlining the correct word(s) in the brackets. [3]

The black lines are caused by light being (**absorbed** / **blocked** / **emitted**) by gases.

The spectrum provides information about the (**size** / **density** / **chemical makeup**) of the object.

The spectrum also provides information about how fast an object is (**expanding** / **using up its fuel** / **moving away from us**).

- (b) The diagram below shows absorption spectra from four different objects in space. The bottom spectrum is from our Sun.



Use the information in the absorption spectra above to answer the following questions. [4]

- (i) State which object is farthest away from Earth.
- (ii) State which object is moving away most slowly from Earth.
- (iii) State **one** conclusion you can make about the chemical makeup of all the objects.
.....
- (iv) State for which object the wavelength of the black lines has increased the most when compared with our Sun.
.....

Section B

Answer **all** questions in the spaces provided.

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Use the information in the separate Resource Folder to answer the following questions.

6. (a) Use the information in **Table 1** to answer the following question.

Tick (✓) the boxes next to the **three** correct statements.

[3]

Saturn orbits the Sun with a velocity of 5 km/s faster than Pluto ☐

Mercury is the hottest planet ☐

All the rocky planets have an atmosphere containing carbon dioxide ☐

Mars and the Earth have the same day length ☐

The Earth has the greatest density ☐

The Moon is the smallest planet ☐

(b) Use the information in **Diagram 3** and your knowledge to answer the following questions.

(i) State how the wavelength of the dark line, λ , for the very distant galaxy is different from the same line in the nearby galaxy. [1]

.....

(ii) State how the frequency of this line for a nearby galaxy compares with the same line in a very distant galaxy. [1]

.....

(iii) State **one** difference about the motion of the very distant galaxy compared to the nearby galaxy. [1]

.....

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(c) Use the information in **Table 1** to answer the following questions.

(i) Europa is an asteroid. Estimate its temperature and orbital period. [2]

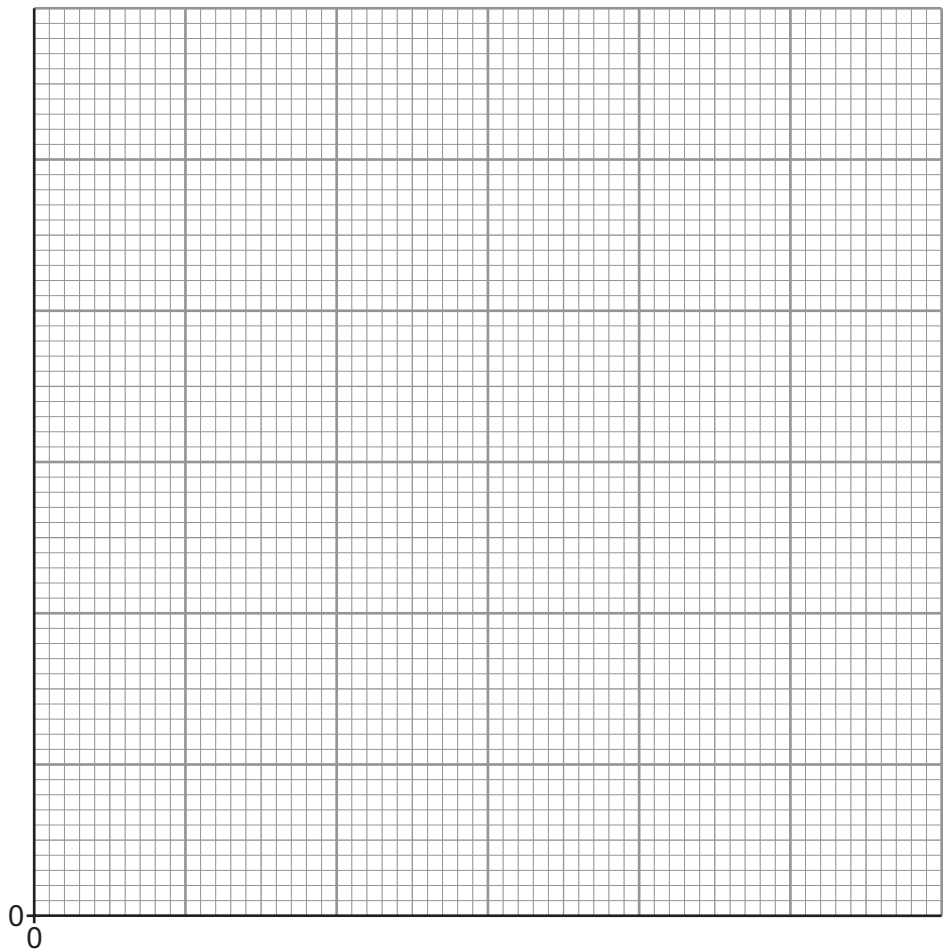
Temperature = °C

Orbital period = days

(ii) Pluto is no longer classed as a planet. State **one** reason why. [1]

(iii) Plot the points on the grid below to show how the orbital velocity of the five planets Mercury, Venus, Earth, Mars and Jupiter depends on distance from the Sun. Draw a suitable line. [4]

Orbital
Velocity
(km/s)



Distance from Sun (AU)

(iv) Orbital velocity is not proportional to the distance from the Sun. Explain how your graph shows this to be true. [2]

.....

.....

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(d) Use your knowledge and the information on **pages 4 and 5** to answer the following questions.

(i) Explain why sunspots appear darker than the surrounding area of the Sun. [2]

.....

.....

(ii) State **two** ways in which human activities may be affected by solar flares. [2]

1.

2.

(iii) Explain why the effect of solar flares on the Earth is not the same for each sunspot cycle. [2]

.....

.....

.....

(e) Use the information in **Diagram 1** to describe the differences between Aristotle’s model of the Solar System and the 2006 model. [4]

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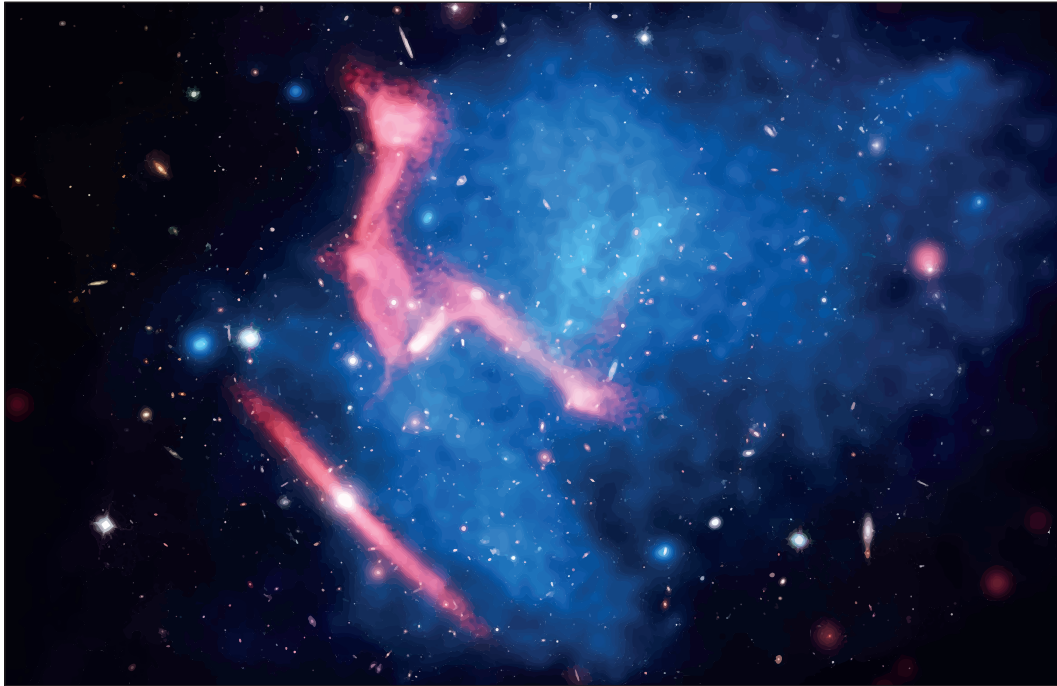
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5. Three images have been combined below to show colliding galaxy clusters 5 billion light years from Earth.

- The background is a Hubble Space Telescope image taken using visible light.
- The blue area is an X-ray image from the Chandra Space Observatory.
- The red area is an image from a radio telescope.



(a) (i) State **one** advantage of using space-based telescopes to produce images of objects in space. [1]

.....

.....

(ii) State **one** disadvantage of using space-based telescopes to produce images of objects in space. [1]

.....

.....

(b) Complete the following sentence by underlining the correct word in the brackets.

It takes radio waves 5 billion (**seconds / hours / days / years**) to travel from the colliding galaxy clusters to reach the Earth. [1]



(c) The image is taken using X-rays, radio waves and visible light.

(i) Which of these waves has the highest frequency? [1]

(ii) Which of these waves has the longest wavelength? [1]

(iii) Name **two** other types of electromagnetic waves that could be used to take images of galaxies. [2]

..... and

(d) The speed of electromagnetic waves through space is 300 000 000 m/s. Some radio telescopes on Earth detect radio waves which have wavelengths between 2 and 10 m.

Use the equation

$$\text{frequency} = \frac{\text{wave speed}}{\text{wavelength}}$$

to calculate the **maximum** frequency of radio waves detected. [2]

frequency = Hz

(e) Images of the Universe can be used to support the current theory of the origin of the Universe.

State the name of this theory. [1]

.....



Section A

Answer **all** questions.

1. Our solar system contains several planets and other objects that orbit the Sun.

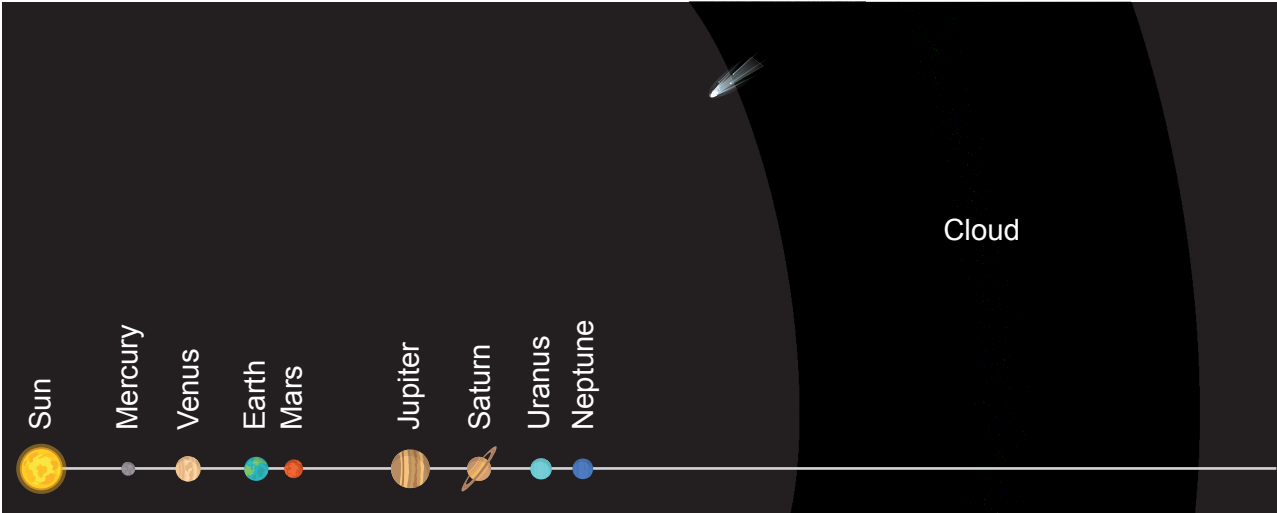


Diagram not drawn to scale

Answer the following questions using your knowledge, the diagram and the words in the box. Each word can be used once, more than once, or not at all.

Comet	Jupiter	Mars	Oort	Venus	Pluto
-------	---------	------	------	-------	-------

- (a) Name the second rocky planet from the Sun. [1]
- (b) Name the gas planet closest to the Sun. [1]
- (c) Name the type of object that orbits the Sun in a highly elliptical orbit. [1]
- (d) Between which **two** planets is the asteroid belt found? [2]
- (e) Name the cloud. [1]



4. (a) Theories about the Universe have changed over time.

Tick (✓) the boxes next to the names of **two** of these theories.

[2]

Liquid State

☐

Steady State

☐

Loud Bang

☐

Big Bang

☐

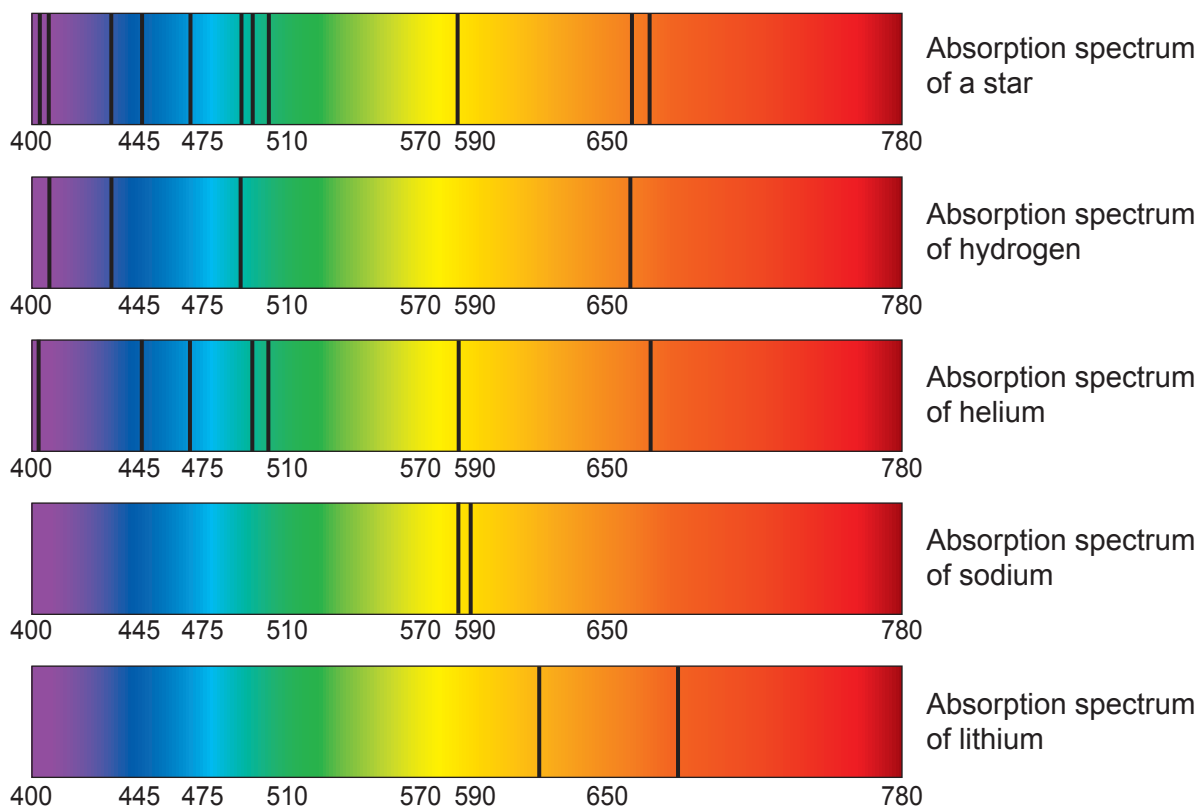
- (b) Images of the Universe are taken using electromagnetic waves.

Name **two** types of electromagnetic wave.

[2]

..... and

- (c) The diagram shows a spectrum from a star and four elements.



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Complete the table by placing a tick (✓) if the element is present in the star. Hydrogen has been done for you.

[1]

Element	Element is present (✓)
hydrogen	✓
helium	
sodium	
lithium	

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(d) The table gives information about the planets in our solar system.

	Mercury	Venus	Earth	Mars	Jupiter	Saturn	Uranus	Neptune
density (kg/m ³)	5427	5243	5514	3933	1326	687	1271	1638
mean distance from the Sun (million km)	58	108	150	228	779	1434	2873	4495
mean speed of orbit (km/s)	47.4	35.0	29.8	24.1	13.1	9.7	6.8	5.4
mean temperature (°C)	167	464	15	−65	−110	−140	−195	−200
number of moons	0	0	1	2	79	82	27	14

Use the information in the table to answer the questions below.

- (i) Name the coldest **rocky** planet. [1]
- (ii) State how many **rocky** planets have moons. [1]
- (iii) Name the gas giant that orbits the Sun with the greatest **speed**. [1]
- (iv) Eris is a dwarf planet. Its orbit is 10 180 million km away from the Sun.
Shaun says it is found in the asteroid belt between the orbits of Mars and Jupiter.
Explain whether you agree with Shaun. [1]

.....

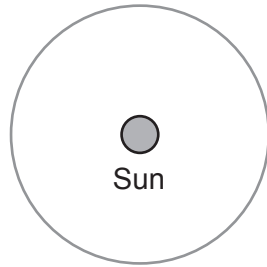
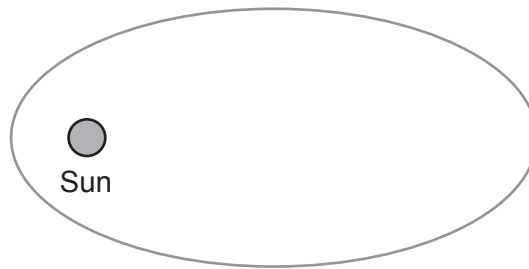
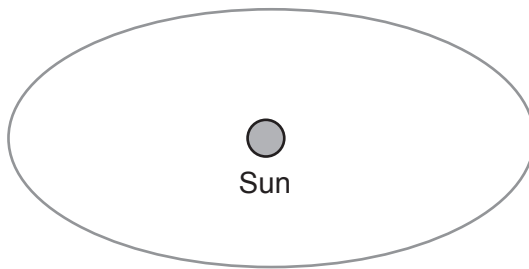
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(e) Tick (✓) the box next to the shape of the orbit of a comet around the Sun.

[1]


☐

☐

☐

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